A 200-Day Trend Filter for 3× Leveraged U.S. Equity ETFs

Reviewer-Response Revision: Financing, Exposure-Matched Controls, Risk Inference, and Reproducibility

Independent quantitative research | Methodological revision v2

12 November 2026

Abstract

This revision reassesses a fixed, long-only rule in which the close of SPY (or QQQ) relative to its 200-session simple moving average (SMA) controls staged next-open investments in the daily-reset 3× product SPXL (or TQQQ). The original observational backtest recorded higher drawdown-adjusted *absolute* losses avoided than unfiltered leveraged buy-and-hold but weaker return and Sharpe performance in the common real-ETF period, 11 February 2010–12 November 2026. We accept the referee's central identification objection: reduced drawdown **by itself** does not demonstrate timing skill because cash exposure is variable. Accordingly, the v2 engine adds unleveraged ETF buy-and-hold, an average-capital-weighted leveraged/cash benchmark, and a lagged-realized-volatility allocator; reconstructs pre-inception leveraged returns with bank-bill funding and consistent adjusted source prices; measures average portfolio weight rather than counting every positive-weight day as a full exposure; and generates regime, rolling-start and trade-ledger diagnostics.

Evidence boundary. The present accompanying historical performance figures are *the uncorrected v1 audit*, not results from the corrected financing and dividend model. Its licensed four-instrument daily source archives cannot be fetched inside the current authoring environment. The corrected model and unit tests are provided, but **new v2 benchmark CAGRs, risk-adjusted comparisons, regime tables, and rolling-start distributions are not asserted as validated empirical findings until a complete source refresh and reproduction run succeeds**. A separate, executable paired circular-block bootstrap operates on the available **201 archived month-end observations** and is labeled explicitly as a coarse-frequency sensitivity audit, not a journal-grade daily-risk inference.

1. Prior art, research design, and falsifiable hypotheses

The 200-day leverage-switching idea is not novel. Gayed and Bilello (2016), *Leverage for the Long Run*, explicitly connect long-term trend, realized volatility, and the decision to employ leverage or move to Treasury bills. Faber (2007) studies moving-average tactical asset allocation. Avellaneda and Zhang (2010) establish the path dependence of leveraged ETF returns, and Moreira and Muir (2017) provide a competing volatility-management mechanism. The contribution here is therefore best characterized as an **implementation-level replication and adversarial benchmark of an existing hypothesis**, not the discovery of SMA-based leverage rotation.

We prestate three distinct hypotheses, none presumed established: (H1) next-open SMA exit reduces severe realized drawdowns relative to 3× buy-and-hold; (H2) it improves a relevant *risk-adjusted metric* versus equal average leveraged exposure and versus a volatility-managed alternative; (H3) findings transport across an S&P 500-linked and a Nasdaq-100-linked product without

asset-specific threshold fitting. H1 is descriptive until inference is performed. H2 is the critical test of market-timing value. SPY and QQQ are correlated U.S. equity exposures; agreement between them is limited transport evidence, not a broad multi-asset replication.

2. The unchanged trading rule

For signal ETF closing adjusted price P_t, define

$$M_t = \frac{1}{200}\sum_{j=0}^{199} P_{t-j}.$$

After close t, a new entry or one additional staged purchase is scheduled if $P_t \geq 1.04M_t$, until targeted weights 20%, 40%, 60%, 80%, and 100% are reached. The target is based on *total portfolio equity at execution*; no automatic sales rebalance an appreciating partial position. Existing shares are retained if $0.97M_t \leq P_t < 1.04M_t$. If invested and $P_t < 0.97M_t$, every leveraged-ETF share is sold. All scheduled orders fill at **open** $t+1$. A uniform assumed 0.10% one-way friction applies per executed notional; taxes and market-impact tails are outside scope. Full stock/ETF purchases permit fractional units.

The 3% filter is a *price-vs-average threshold*, not a fixed 3% stop on ETF equity. A stretched trend can be far above its own SMA; therefore the rule can still suffer large and rapid leveraged drawdowns before exiting.

3. Correcting cash, financing, and distributions

3.1 Historical T-bill proxy and safe information timing

The revision uses monthly *last available* daily quotations from the Federal Reserve's DTB3 3-month Treasury bill bank-discount series. Only **the previous calendar month's** last available quote is used during month m: the current month's final observation is never used prospectively inside that month. As DTB3 quotes on a bank-discount, 360-day convention, a 91-day investment-equivalent annual rate is approximated by

$$\tilde{r}_{f,t} = \frac{(365/360)d_{m-1}}{1-(91/360)d_{m-1}},$$

where d_{m-1} is the prior-month-end quoted annual discount yield in decimal form. This is a stylized bill reinvestment proxy, not an exact tradeable return series; actual cash accounts may receive less and may have account-specific restrictions. Idle portfolio cash compounds at approximately $1+\tilde{r}_{f,t}/252$ per trading day. The machine-readable derivative contains 335 monthly observations spanning December 1998–October 2026; October's latest sourced quote was 26 October. The isolated negative September 2015 quote (−0.01% annual bank discount) is explicitly preserved in the rate decoder rather than silently rounded to zero.

3.2 Pre-inception leveraged return, using consistent total-return underlying

Let r_t^{TR} be the unleveraged ETF daily **adjusted total-return proxy**. The pre-inception synthetic 3× closing series follows

$$L_t = L_{t-1}\left(1 + 3r_t^{TR} - 2\frac{\tilde{r}_{f,t} + s}{252} - \frac{f}{252}\right),$$

where $s \geq 0$ is an explicit additional annual financing spread **per borrowed unit** and $f = 0.0095$ is an illustrative annual operating fee. Two funded units are charged at the short-rate proxy, unlike the prior v1 code, which omitted the borrowing term. The financing and fee are not charged a second time for actual SPXL/TQQQ prices, which already reflect the funds' realized economics. Synthetic openings model 3× the overnight underlying move and an illustrative half-day carry; they remain **not executable historical quotes**. Exact swap costs, futures basis, ETF tracking and historical distribution implementation are not observed and hence cannot be recovered from a price file alone.

SPY and SPXL source files have `Adj Close`; the adjusted Open is reconstructed as `Open × Adj Close / Close` on each date. The already-auto-adjusted QQQ/TQQQ inputs remain adjusted. The recent tail must use matching adjustment conventions. Source stitching **fails rather than rescales** if the archive and the independent overlap close differ by more than one basis point. The IBKR-only snapshot is retained solely as a dated, partly price-return, *legacy comparison*; any new corrected total-return analysis uses vendor-consistent input or stops with an error.

3.3 Economic intuition and its limitations

For ideal continuous-time constant daily-reset leverage ℓ, arithmetic underlying drift μ, variance σ^2, and a funding rate r_f, the approximate leveraged log growth is

$$g_\ell \simeq \ell\mu - \frac{\ell^2\sigma^2}{2} - (\ell - 1)r_f - f.$$

At $\ell = 3$, the variance penalty is $4.5\sigma^2$, approximately 55.1% annually for $\sigma = 35\%$, versus 6.48% for $\sigma = 12\%$, holding all else fixed. This illustrates why high-volatility regimes can damage compounded leveraged growth. It is **not evidence that an SMA filter actually predicts future volatility or return**. The v2 `regime_nextday.csv` generator therefore sorts observations by the close-of-day signal regime, then computes the *following* session's underlying mean return, volatility, and illustrative leveraged growth, avoiding same-bar contemporaneous circular reasoning.

4. Independent controls that determine timing value

The experiment explicitly compares five investment rules on common timestamps and identical data:

Method	Target weight in 3× ETF	Cash treatment	Timing information
SMA200 / +4% / −3%	Original five-stage state machine	Prior-month T-bill yield	Close t, open $t+1$
Leveraged ETF buy/hold	100% after first open	None except residual	None
Unleveraged SPY or QQQ buy/hold	100% in 1× ETF	None except residual	None

Method	Target weight in 3× ETF	Cash treatment	Timing information
Exposure-matched	Constant leveraged allocation equal to **realized mean SMA weight**, remainder cash; rebalance every 21 trading sessions	Prior-month T-bill yield	Retrospective diagnostic only
Volatility-managed	Weight $\min(1, 0.30/\hat{\sigma}_{63,t})$, 63-day lagged-realized ETF volatility; rebalance every 5 sessions	Prior-month T-bill yield	Trailing through close t

The **average weight** is $\bar{w} = T^{-1}\sum_t w_t$ where $w_t = \text{shares}_t\,\text{price}_t/\text{NAV}_t$; a 20%-weight day is therefore only one-fifth of a full-risk day, not 100% exposed. The exposure-matched control calibrates to a **full-sample realized mean SMA weight** and thus is **ex-post only**, not a prospective investment rule; it isolates how much lower average exposure explains. The 30% volatility target is a fixed, plainly stated diagnostic choice, not fit on these two funds. Any claim of superior SMA *timing* demands a relevant advantage versus these alternatives with appropriately calibrated risk and statistical uncertainty, not just superiority to 100%-exposed leveraged ETF drawdown.

5. Results audit and Calmar correction

The following **v1** figures are preserved to show what was criticized. They are *not* asserted as corrected total-return, financing-inclusive v2 results. The common actual-ETF dates were 11 February 2010–12 November 2026, with 10 bp execution cost and **zero** yield on uninvested cash.

Original 2010–2026 audit	SMA CAGR	3× hold CAGR	SMA max DD	Hold max DD	SMA Calmar	Hold Calmar
SPXL	21.14%	28.94%	−52.88%	−76.86%	0.400	0.377
TQQQ	26.93%	40.91%	−62.09%	−81.66%	0.434	**0.501**

The TQQQ SMA rule underperformed unfiltered leveraged buy/hold on CAGR, Sharpe, **and Calmar**, and only offered a smaller *absolute* observed maximum drawdown. Even SPXL's Calmar advantage is narrow. Thus the original draft's blanket risk-management framing was too strong. Since lower exposure plausibly accounts for some downside improvement, **the independent value of the SMA filter remains unproven** until the v2 control runs finish.

6. Paired risk uncertainty: newly computed, frequency-limited evidence

A paired fixed-length **circular block bootstrap** reuses the archived *201 monthly close-to-close observations* of both strategies and their leveraged buy/hold controls (5,000 draws, 12-month blocks, seed 20261112). Each bootstrap draw uses the **same sampled month indices** for the strategy and its control, preserving contemporaneous co-movement. These are percentile intervals from an explicitly *retrospective* sample; the method is not the Politis–Romano stationary bootstrap, does not use Ledoit–Wolf's complete studentized Sharpe procedure, and does not correct for earlier parameter search.

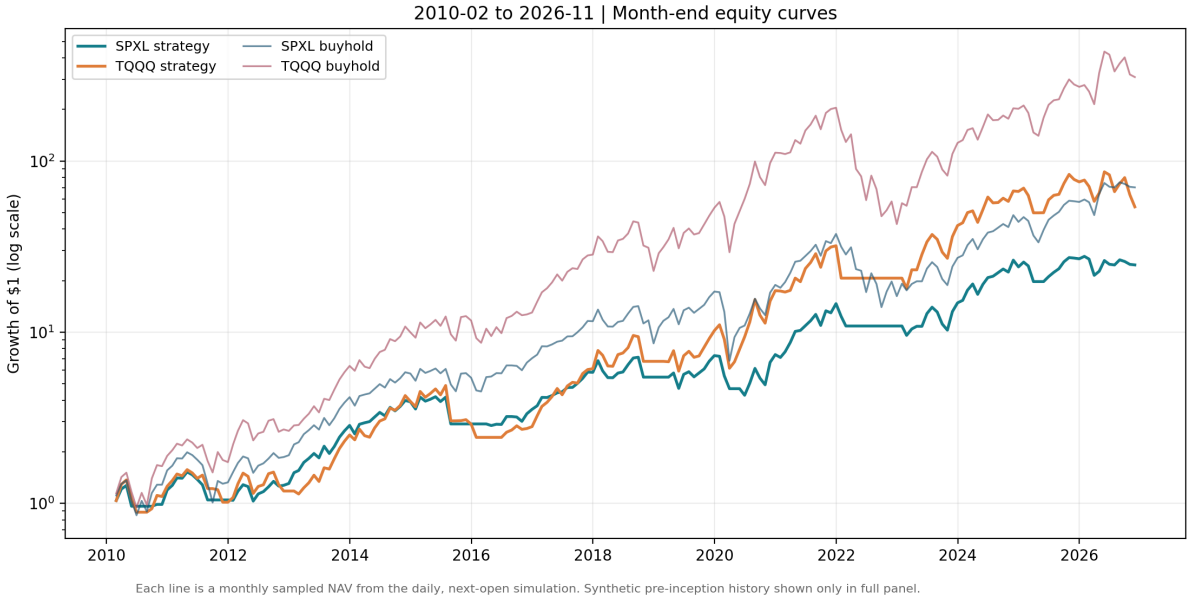


Figure 1: Figure 1. Legacy real-ETF 2010–2026 audited equity curves, *before* funding, dividend and benchmark revisions. Do not read as v2 performance.

Asset	Risk difference (strategy minus 3× hold)	Observed	Paired 95% percentile interval
SPXL	Monthly-return Sharpe (rf = 0)	−0.047	[−0.276, +0.184]
TQQQ	Monthly-return Sharpe (rf = 0)	−0.121	[−0.375, +0.182]
SPXL	Max drawdown, sampled monthly NAV	+0.213	[−0.066, +0.338]
TQQQ	Max drawdown, sampled monthly NAV	+0.289	[−0.215, +0.351]
SPXL	Worst 5% monthly CVaR	+0.052	[−0.009, +0.109]
TQQQ	Worst 5% monthly CVaR	+0.025	[−0.029, +0.075]

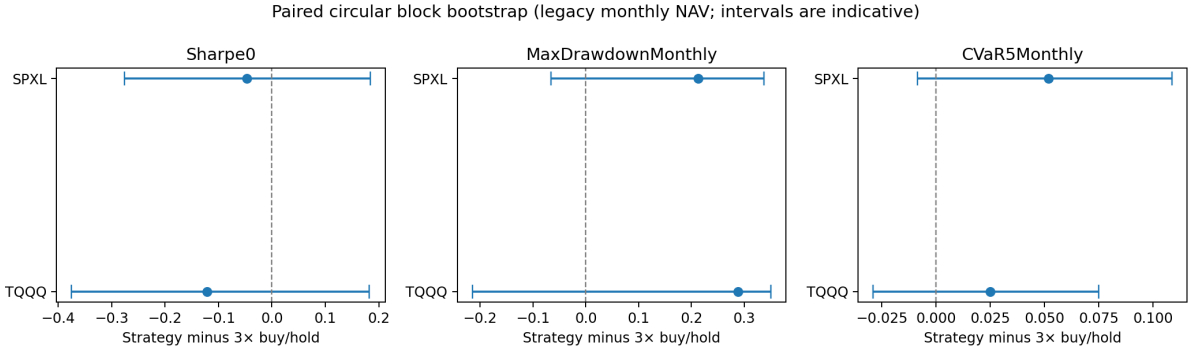


Figure 2: Figure 2. Legacy monthly paired bootstrap intervals. Every shown interval intersects zero; these do not estimate daily policy-resampling uncertainty.

The sign of drawdown/CVaR differences is written so a positive value indicates a **smaller absolute loss** under SMA. All six exploratory intervals cross zero. This does not demonstrate risk-indicator superiority. Month-end sampling misses adverse daily peaks and therefore these figures cannot be substituted for daily-sampled drawdown inferential results. A valid future analy-

The program additionally writes full-sample *daily* risk summaries (mean/conditional mean drawdown, CDaR10, CVaR5, Ulcer Index, underwater fraction, longest underwater spell), a trade ledger mapping each full exit to the subsequent first-stage reentry, and a monthly-start distribution of strategy CAGR and maximum drawdown. The ledger records the leveraged ETF return **forgone between exit and reentry**. A positive foregone return is a reentry opportunity cost; a negative one reflects avoiding part of a decline. It is not counterfactual causal identification, and the trade count is small. These features are implemented but do not have populated corrected-run tables in this release because the four raw price streams are unavailable locally.

Mechanical weakness to test next: as price rises far above a trailing average, the absolute exit distance expands; during violent crashes a next-open execution can suffer a material gap. Alternative volatility-scaled buffers or trade-level drawdown stops can be prospectively preregistered as **new** experiments, not silently substituted into this baseline.

8. Code validation, data requirements, and pending publication gates

The revised code includes offline unit tests verifying next-open timing, staged orders, proper liquidation, average capital weight, interest accrual on uninvested cash, last-month-only Treasury quotations, failure on an unaligned vendor splice, and that higher financing costs worsen the synthetic invested return. A separate paired monthly-risk file is generated directly from shipped derived NAV snapshots and runs entirely offline. The entire corrected cross-asset backtest requires four *locally retrieved* adjusted OHLC archives plus a tail that passes the 1 bp overlap audit.

The following **must** be satisfied before the paper can honestly be treated as a revised empirical replication rather than an executable research protocol:

1. Acquire licensed, consistent total-return OHLC for all four instruments, with dividend/split reconciliation and independent checks of official fund NAV total returns.
2. Rerun `scripts/review_revision.py`; publish the **new** v2 results rather than reusing v1 metrics; record the source file hashes and merge-status report.
3. Evaluate 1×, cash-weight-matched, and volatility-managed benchmarks on identical windows, with both matched-weight and volatility-target opportunity costs explicitly discussed.
4. Run a daily dependent risk bootstrap that recalculates strategy decisions on resampled price *paths*, and a truly future frozen-parameter paper-trading sample.
5. Release any additional parameter trials together with their outcomes. Do not describe the 2018–2026 split as an untouched holdout given the historical experimentation.

The record remains *exploratory* until these conditions are met. This is a deliberately stronger and more falsifiable characterization than “risk management validated.”

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